Organic & Biomolecular Chemistry

REVIEW

View Article Online
View Journal | View Issue

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

Check for updates

Cite this: *Org. Biomol. Chem.*, 2019, **17**, 1027

Biosynthesis of bioactive natural products from Basidiomycota

Hsiao-Ching Lin, *[a,b,c] Ranuka T. Hewage, [b,d] Yuan-Chun Lu [a,c] and Yit-Heng Chooi [e]

Received 7th November 2018,
Accepted 13th December 2018

DOI: 10.1039/c8ob02774a

rsc.li/obc

The Basidiomycota, also called club fungi, comprise a diverse group of fungi. Basidiomycota are strongly related to ecosystem functioning along with human life. These fungi display a wide range of bioactivities, and some are known to produce of deadly toxins or hallucinogens. Some Basidiomycota have be used as medicinal mushrooms for thousands of years. Recently, the biosyntheses of several classes of natural products from Basidiomycota have been reported. Here, we review recent studies on the biosynthetic pathways and enzymes of bioactive natural products from Basidiomycota fungi, with a focus on terpenoids, alkaloids, ribosomally synthesized and post-translationally modified peptides (RiPPs), and polyketides.

[a]*Institute of Biological Chemistry, Academia Sinica, Taipei 115, Taiwan, People's Republic of China. E-mail: hsiaoching@gate.sinica.edu.tw*
[b]*Chemical Biology and Molecular Biophysics Program, Taiwan International Graduate Program, Academia Sinica, Taipei 115, Taiwan, People's Republic of China*
[c]*Institute of Biochemical Sciences, National Taiwan University, Taipei 106, Taiwan, People's Republic of China*
[d]*Department of Chemistry, National Taiwan University, Taipei 106, Taiwan, People's Republic of China*
[e]*School of Molecular Sciences, The University of Western Australia, Perth, WA 6009, Australia*

1. Introduction

Fungi are abundant worldwide and show great diversity. Basidiomycota together with Ascomycota represent one of the two major phyla in the fungal kingdom. The basidia are the characteristic cellular structure of Basidiomycota and produce haploid sexual spores known as basidiospores. Unlike Ascomycota, which are usually monokaryotic for most of their lives, Basidiomycota have long-lived dikaryotic states. For thousands of years, many Basidiomycota fungi have been used for

Hsiao-Ching Lin

Hsiao-Ching Lin obtained her Ph.D. in natural product chemistry in 2011 from the Department of Pharmaceutical Science, National Taiwan University under the supervision of Professor Shoei-Sheng Lee. She moved to University of California, Los Angeles in 2012 for a postdoctoral position to study the biosynthesis of fungal secondary metabolites with Professor Yi Tang. In 2015, she began her independent academic career as an assistant research fellow at Institute of Biological Chemistry, Academia Sinica, Taiwan, where she is working on natural product biosynthesis.

Ranuka T. Hewage

Ranuka T. Hewage received his B.S. degree in Chemistry from the University of Sri Jayewardenepura and then joined the Medical Research Institute, Sri Lanka in 2011. In 2014, he received a M.S. degree in Chemical Biology with a focus on natural products chemistry from Chulabhorn Graduate Institute, Thailand (Now Chulabhorn Royal Academy) under the supervision of Professor Prasat Kittakoop. In 2016, he received a Taiwan International Graduate Program scholarship to pursue a Ph.D. from the Department of Chemistry, National Taiwan University. He is currently a Ph.D. student in Hsiao-Ching Lin's group at the Institute of Biological Chemistry, Academia Sinica, Taiwan.

This journal is © The Royal Society of Chemistry 2019

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

culinary purposes, including shiitake (*Lentinula edodes*), and for medicinal purposes, such as Ling Zhi (*Ganoderma lucidum*) and Fu Lin (*Wolfiporia cocos*). In contrast, some Basidiomycota are poisonous, such as the death cap (*Amanita phalloides*). Numerous natural products have been characterized from cultures or fruiting bodies of Basidiomycota.[1,2] A considerable number of compounds with structures distinct from those produced by other microbes have been reported, highlighting the ability of Basidiomycota to utilize novel biocatalysts as chemical tools for the biosynthesis of natural products. Compared to other microbial producers such as bacteria and filamentous fungi, especially Ascomycota, Basidiomycota fungi remain understudied for the discovery and application of enzymatic machinery in biosynthesis. With the increase in the number of genome sequence data available for Basdiomycota in recent years,[3] the biosyntheses of several classes of natural products from this phylum have been reported. Basidiomycota fungi are prolific producers of terpenoids, especially sesquiterpenoids, diterpenoids and triterpenoids. The biosynthesis of sesquiterpenoids from Basidiomycota, including the sesquiterpene synthases and their cyclization mechanisms, has recently been covered in several excellent reviews.[4–6] New classes of sesquiterpene synthases continue to be discovered. For example, Flynn *et al.* reported a hirsutene synthase from *Stereum hirsutum* with an unexpected fused C-terminal 3-hydroxy-3-methylglutaryl-coenzyme A synthase domain.[7] This review does not cover sesquiterpenoids; instead, it focuses on the recently discovered biosynthetic pathways of bioactive natural products in the classes of diterpenoids, triterpenoids, alkaloids, ribosomally synthesized and post-translationally modified peptides (RiPPs) and polyketides from Basidiomycota. Specifically, this review focuses on the functions of the biosynthetic enzymes discovered in these pathways.

2. Biosynthesis of pleuromutilin

Pleuromutilin is an antibacterial diterpene that consists of a "propellane-like" 5–6–8-tricyclic hydrocarbon skeleton and a glycolic acid side chain (Fig. 1A). Pleuromutilin was first isolated from the fungi *Pleurotus mutilus* and *Pleurotus passeckerianus* (now *Clitopilus passeckerianus*) in 1951.[8] It exhibits antibacterial activity against Gram-positive pathogens and mycoplasma species by selectively inhibiting protein synthesis through interaction with prokaryotic ribosomes.[9] The semi-synthetic derivatives tiamulin and valnemulin (Fig. 1B) have been produced for veterinary use, particularly for the prophylaxis of dysentery, pneumonia, and mycoplasmal infections in swine and poultry.[10] Retapamulin (Fig. 1B), another semi-synthetic derivative, is used as topical antibiotic to treat impetigo and infected small lacerations, abrasions and sutured wounds in humans.[11]

The biosynthesis of pleuromutilin has been studied by isotope feeding.[9,12,13] Recently, the *ple* gene clusters of pleuromutilin from both *Clitopilus pseudopinsitus* and *P. passeckerianus* were characterized.[14–16] Both contained seven genes responsible for pleuromutilin biosynthesis, including a GGPP synthase (*Pl-ggs*), a terpene synthase (*Pl-cyc*), three cytochrome P450s (*Pl-p450-1*, *Pl-pe450-2*, and *Pl-p450-3*), a short-chain dehydrogenase/reductase (*Pl-sdr*), and an acyltransferase (*Pl-atf*), respectively termed *ple4*, *ple3*, *ple5*, *ple6*, *ple1*, *ple7*, and *ple2* in *C. pseudopinsitus*. The function of each gene was characterized with stepwise heterologous expression in *Aspergillus oryzae*. Pl-cyc/ple3 is a bifunctional diterpene synthase (DTS) that produces tricyclic premutilin. Two cytochrome P450s, Pl-p450-1/ple5 and Pl-p450-2/ple6, catalyze the stereospecific hydroxylation of C-11 and C-3 on premutilin, respectively. Pl-sdr/ple7 converts the 3-hydroxyl group into a ketone functionality and forms mutilin, and Pl-atf/ple2 then

Yuan-Chun Lu

Yuan-Chun Lu is currently a master's student under the supervision of Dr Hsiao-Ching Lin at the Institute of Biochemical Sciences, National Taiwan University. She holds a B.S. degree in the Department of Biochemical Sciences and Technology, National Taiwan University, Taiwan. Her current research interest is the biosynthesis of natural products in Basidiomycota.

Yit-Heng Chooi

Yit-Heng Chooi received his Ph.D. in 2009 from RMIT University, Australia and moved to the USA for a postdoctoral position at the University of California, Los Angeles with Prof. Yi Tang. He returned to Australia for an Australian Research Council (ARC) Discovery Early Career Researcher Award in 2013 at the Australian National University and started his own group at the University of Western Australia in mid-2015. He was awarded an ARC Future Fellowship in 2016. His research interests are in genome-based natural product discovery and biosynthesis.

 This journal is © The Royal Society of Chemistry 2019

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

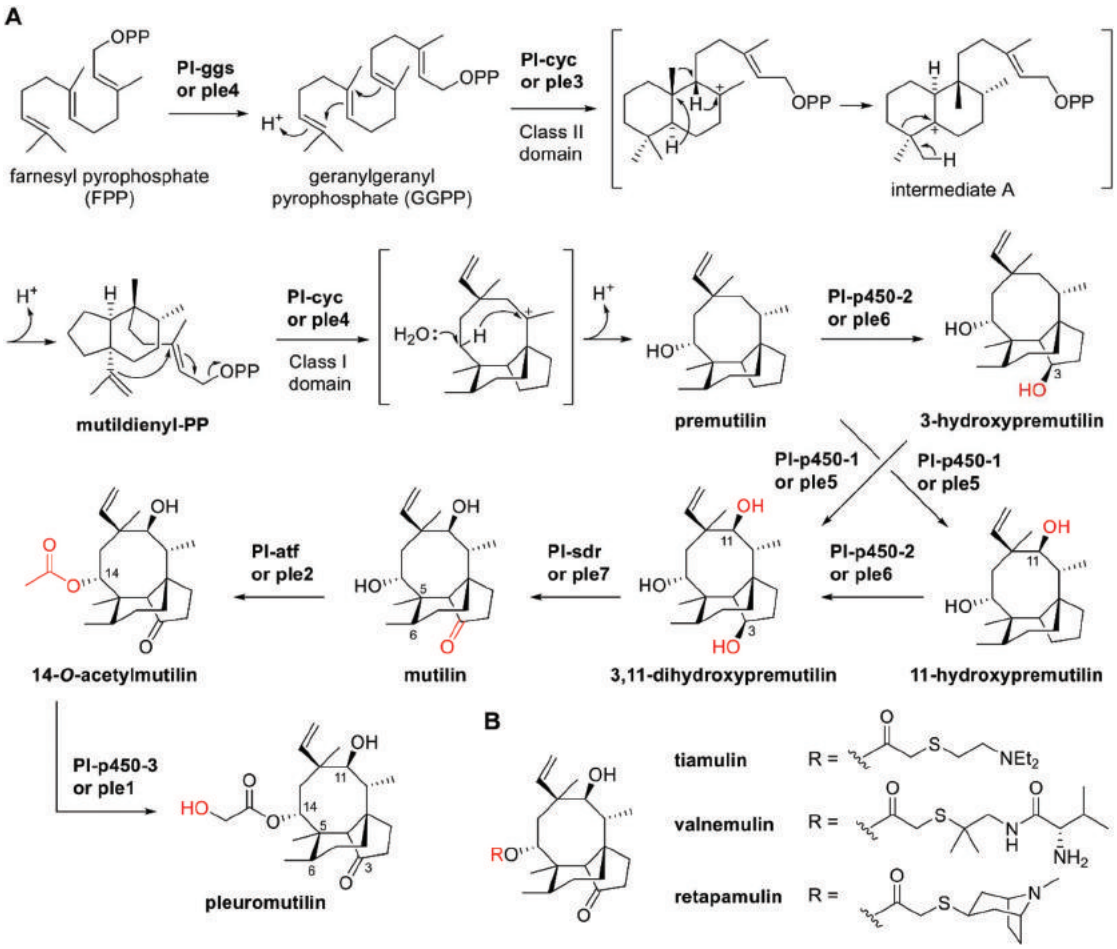

Fig. 1 (A) Biosynthesis of pleuromutilin. (B) Structures of the clinical medicines tiamulin, valnemulin and retapamulin.

transfers an acetyl group to 14-OH. The final step is catalyzed by Pl-p450-3/ple1, which hydroxylates the α-position of the acetyl side chain to form pleuromutilin (Fig. 1A).

Pl-cyc/ple3 is a unique DTS that shows novel Class II DTS activity. It catalyzes ring contraction and forms a 5–6-bicyclic ring system, which is different from the *trans*-decalin core ring catalyzed by other Class II DTSs in labdane-related diterpenoid biosynthesis. ple3/Pl-cyc contains two catalytic domains, an N-terminal class II DTS domain with a $Dx^{311}DM$ motif that catalyzes the protonation-initiated cyclization, and a C-terminal class I DTS domain with a $^{649}DDxxD$ motif that catalyzes ionization-dependent dephosphorylation and further cyclization. The mutations of D311A and D649A on ple3 abolish the production of premutilin when incubated with GGPP *in vitro*. The incubation of both mutants together with GGPP restored the cyclization activity, indicating the importance of the aspartate-rich motifs for the catalysis of cyclization by Ple3.

For the cyclization mechanism of Pl-cyc/ple3, the Class II domain catalyzes the initial bicyclization of GGPP following a series of 1,2-shifts to yield the reaction intermediate A with a decalin skeleton. Subsequently, A is converted into mutildienyl-PP, which can be isolated from the CpPS:D649L mutant, by ring contraction with deprotonation of the methyl group on the isopropyl substituent.[17] Mutildienyl-PP is further cyclized by the Class I domain in Pl-cyc/ple3 following a 1,5-hydride shift and the addition of water to form premutilin (Fig. 1A).

3. Biosynthesis of cyathane diterpenoid

Cyathane diterpenoids, which contain characteristic 5–6–7 tricyclic skeletons, have been isolated from a variety of bacteria, fungi, and fruiting plants.[18] They displayed a wide variety of biological activities such as antibiotic, antimicrobial, and anticancer activities, along with the ability to promote the synthesis of nerve growth factor (NGF). Cyathin A_3 and allocyathin B_3 (Fig. 2A) were the first cyathane diterpenoids to be isolated and structurally characterized from the bird's nest fungus *Cyathus helenae* in early 1970s.[19] Erinacines, from the mushroom *Hericium erinaceurm*, are glycosylated analogues of allocyathin B_2. Erinacines have been reported to exhibit considerable potency in stimulating the synthesis of NGF and show

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

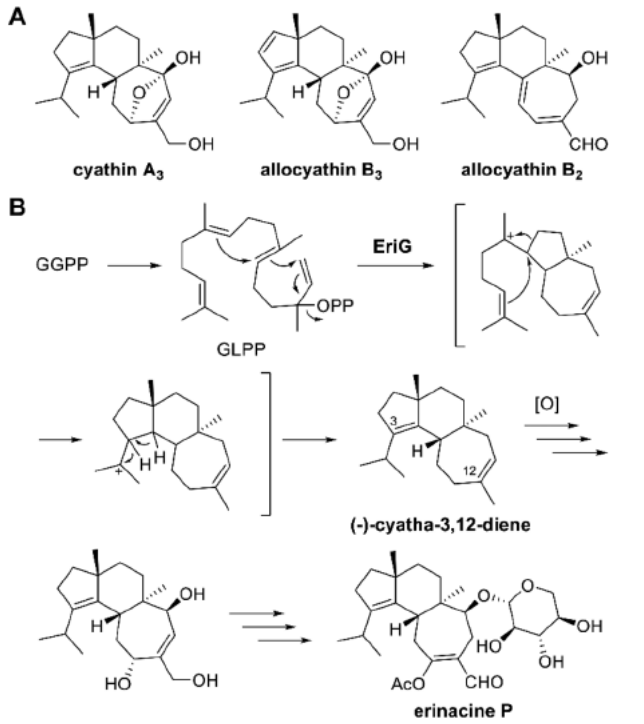

Fig. 2 (A) Structures of the cyathane diterpenoids cyathin A_3, allocyathin B_3 and allocyathin B_2. (B) Biosynthesis of the cyathane diterpenoid erinacine P. EriG is a membrane-bound diterpene cyclase that converts GGPP into (−)-cyatha-3,12-diene. (GGPP: geranylgeranyl diphosphate; GLPP: geranyllinalyl diphosphate).

potential as a therapeutic agent to treat neurodegenerative disease (Fig. 2).[20]

In the biosynthesis of erinacines, the *Eri* gene cluster from *H. erinaceurm* has been characterized and contains one GGPPS (*EriE*), two annotated UbiA prenyltransferases (*EriF* and *EriG*), a UDP-glycosyltransferase (*EriJ*), and three cytochrome P450s (*EriA*, *EriC*, and *EriI*).[21] EriG is a membrane-bound diterpene cyclase that converts GGPP into (−)-cyatha-3,12-diene; no product was observed with EriF under the same experimental conditions (Fig. 2B). The activity of EriG was confirmed by *in vitro* assay with an *Escherichia coli* BL21 (DE3) membrane fraction containing EriG along with the structural characterization of (−)-cyatha-3,12-diene isolated from the large-scale culture of GGPP-engineered *E. coli* BW25 113 expressing EriG. The rest of the aforementioned genes were proposed to be involved in the biosynthesis of erinacine P based on the transcription level of the genes and metabolic profiles of erinacine P-producing and non-producing conditions. However, their functions still need to be characterized with further experiments.

Two enzymes, FmaTC and PtmT1, that belong to UbiA superfamily have been reported as membrane-bound terpene cyclases from Ascomycota and bacteria, respectively. FmaTC from *Aspergillus fumigatus* is a Class I sesquiterpene cyclase that catalyzes the conversion of farnesyl pyrophosphate into β-*trans*-bergamotene in fumagillin biosynthesis.[22] PtmT1 from *Streptomyces platensis*, which is a Class I diterpene cyclase, catalyzes the formation of *ent*-atiserene from *ent*-copalyl diphosphate in platencin biosynthesis.[23] Like FmaTC, EriG contains two highly conserved motifs, NXXX(G/A)XXXD and DXXXD, corresponding to motif I and motif II of UbiA, respectively. Thus, EriG is a member of the recently discovered UbiA-like membrane-bound Class I diterpene cyclases. It will be interesting to further investigate whether these membrane-bound Class I terpene cyclases evolved convergently from UbiA-like ancestors in Ascomycota and Basidiomycota in fungi and in bacteria, or if it is a case of lateral transfer.

4. Biosynthesis of triterpenoids

Basidiomycota are prolific producers of bioactive triterpenoids, such as the cytotoxic ganoderic acid T from *Ganoderma lucidum*[24] and antcin B from *Antrodia cinnamomea*[25] and the cytotoxic or apoptotic lanosterane-type triterpenoids from the genera *Innonotus*, *Poria*, *Laetiporus*, and *Daedalea*.[26] *G. lucidum*, called Lingzhi in Chinese, has been used in traditional Chinese medicine for the promotion of longevity and maintenance of vitality. *G. lucidum* is probably the most well studied species among the triterpenoid-producing fungi. More than 250 triterpenoids have been identified from this species.[24] Nonetheless, compared to triterpenoid biosynthesis in plants, which is relatively well characterized, less is known about the biosynthesis of triterpenoids in Basidiomycota.

In fungi, triterpenoids are derived from the cyclization of oxidosqualene into lanosterol, which is also the precursor to ergosterol.[27] Multiple oxidation steps, potentially catalyzed by a number of cytochrome P450 monooxygenases, modify the triterpene skeleton to generate diverse triterpenoid metabolites. In Basidiomycota, recent genomic and transcriptomic data show that triterpene cyclases do not appear to be clustered with the genes encoding downstream tailoring enzymes; instead, they appear to be co-regulated together at some stages of tissue development. In the medicinal fungus *G. lucidum*, 78 cytochrome P450 genes out of 197 were shown to be co-expressed with the lanosterol synthase gene and were upregulated in the transition from mycelia to primordia.[28]

The first cytochrome P450 monooxygenase, CYP5150L8, was recently characterized in relation to the biosynthesis of ganoderic acid from *G. lucidum*.[29] CYP5150L8 catalyzes the successive, three-step C-26 oxidation of lanosterol into 3-hydroxy-lanosta-8,24-dien-26-oic acid (HLDOA) through intermediates 3-hydroxy-lanosta-8,24-dien-26-ol (HLDO) and 3-hydroxy-lanosta-8,24-dien-26-al (HLDA; Fig. 3). Its function was reconstituted in *S. cerevisiae* in both *in vivo* and *in vitro* experiments.

5. Biosynthesis of psilocybin

Mushrooms containing the hallucinogenic alkaloids psilocybin and psilocin (Fig. 4) have long been used for medicinal

 This journal is © The Royal Society of Chemistry 2019

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

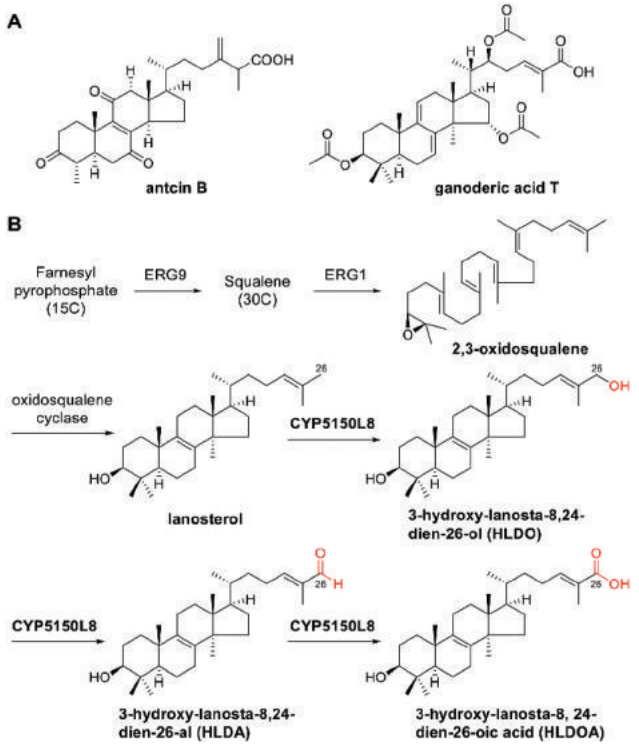

Fig. 3 (A) Structures of antcin B and ganoderic acid T. (B) Biosynthesis of HLDOA, a ganoderic acid from *G. lucidum*.

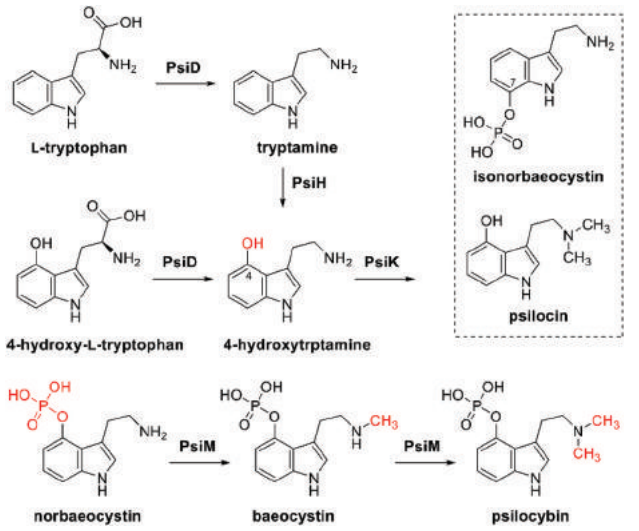

Fig. 4 Biosynthesis of psilocybin in *P. cubensis* and *P. cyanescens*.

and religious purposes by the native people of Mexico.[30] The genus *Psilocybe* consists of the so-called "magic mushrooms," which are best known for their psychedelic properties, which result in altered perception when ingested. Psilocybin and psilocin were first identified as the psychoactive compounds in *Psilocybe* mushrooms in 1958 by Albert Hofmann.[31,32] Psilocybin is the main constituent from this fungal genus and acts as a prodrug. It is rapidly dephosphorylated to generate the actual hallucinogen, psilocin, which is an agonist with high affinity for the human 5-HT_{2A} receptor in the brain.[33]

The biosynthetic pathway of psilocybin was characterized by Fricke *et al.*[34] Psilocybin biosynthetic gene clusters (*psi*) have been identified from *Psilocybe cubensis* and *Psilocybe cyanescens*. Both clusters contain genes encoding enzymes with a high degree of sequence identity (between 78% and 85%). Four psilocybin biosynthesis enzymes have been characterized, including a new class of fungal L-tryptophan decarboxylases (PsiD), a kinase (PsiK), a *S*-adenosyl-l-methionine (SAM)-dependent *N*-methyltransferase (PsiM), and a P450 monooxygenase (PsiH). L-Tryptophan is first decarboxylated to give tryptamine with PsiD, which belongs to the PLP-independent phosphatidylserine decarboxylase family (E.C. 4.1.1.65), but is not related to PLP-dependent aromatic L-amino acid decarboxylases. Interestingly, PsiD also accepts 4-hydroxy-L-tryptophan as a substrate for decarboxylation.

Subsequent modification reactions are mediated by PsiH, PsiK and PsiM in a virtually linear process to biosynthesize psilocybin (Fig. 4). The PsiH monooxygenase catalyzes 4-hydroxylation on tryptamine to afford 4-hydroxyltryptamine but does not accept L-tryptophan for 4-hydroxylation. PsiK, which belongs to the 5-methylthioribose family of small-molecule kinases, catalyzes 4-*O*-phosphorylation and converts 4-hydroxyltryptamine into norbaeocystin, a demethylated analogue of psilocybin. It was noted that a decarboxylated substrate is required by PsiK kinase. In the final reactions, PsiM methyltransferase, which only recognizes phosphoryloxy groups on substrates, catalyzes iterative *N*-methyl transfer to install the *N*,*N*-dimethyl group, producing psilocybin. Psilocybin was enzymatically synthesized *in vitro* from 4-hydroxyltryptophan in a PsiD/PsiK/PsiM in a one-pot reaction.[34] In a recent study on biocatalytic production, the addition of TrpB, a tryptophan synthase from *P. cubensis*, resulted in the formation of psilocybin from 4-hydroxyindole and L-serine through the enzymatic reactions of TrpB/PsiD/PsiK/PsiM. An unnatural product 7-phosphoryloxytryptamine (isonorbaeocystin) was obtained from the *in vitro* reaction by adding substrate 7-hydroxyltryptamine.[35]

6. Biosynthesis of amatoxins and phallotoxins

The amatoxins and phallotoxins are two major toxic families from the poisonous mushrooms in genus *Amanita* section Phalloideae.[36] In 1937, Feodor Lynen and Ulrich Wieland in Heinrich Wieland's laboratory identified the major rapid toxin phalloidin from *Amanita phalloides*, which is commonly known as the "death cap."[37] The major slow-acting toxin α-amanitin was later identified in 1941. These two toxic families show mammalian toxicity with different modes of action. Amatoxins inhibit RNA polymerase II after cellular uptake, and the human LD_{50} for α-amanitin is ~0.1 mg kg^{-1}. Phallotoxins bind to and stabilize F-actin and are poisonous when injected because of poor oral absorption. Naturally

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

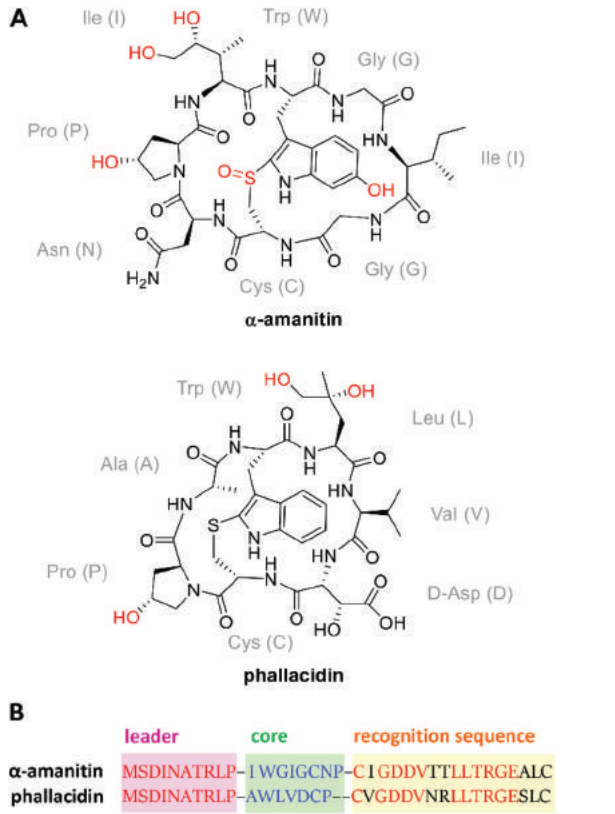

Fig. 5 (A) Structures of α-amanitin, a bicyclic octapeptide, and phallacidin, a bicyclic heptapeptide. (B) Alignment of the AMA1 and PHA1 precursor peptides, illustrating the highly conserved leader peptide and recognition sequence in red.

occurring amatoxins and phallotoxins are bicyclic octapeptides and bicyclic heptapeptides, respectively (Fig. 5A). They both contain the following structural features: (i) tryptathionine, which is formed by cross-bridging between the thiol of Cys to the C2 position of Trp and is oxidized to a (*R*)-sulfoxide in amatoxins and remains a thioether in phallotoxins; and (ii) 4-hydroxy-L-Pro with *trans* configuration (2*S*,4*R*) in amatoxins and *cis* configuration (2*S*,4*S*) (D-amino acid) in phallotoxins. Post-hydroxylation modifications and differences in amino acid residues (Asn^7 or Asp^7 for amatoxins and Ala^4-Thr^5 or Val^4-Asp^5 for phallotoxins) on their bicyclic peptide skeletons lead to the ten known amatoxins and seven known phallotoxins.

The biosyntheses of α-amanitin, an amatoxin, and phallacidin, a phallotoxin, begin with a ribosomally synthesized precursor peptide containing 35 or 34 amino acids, respectively (Fig. 5B).[38,39] The precursor peptide contains a 10-amino-acid leader peptide; a 7- or 8-amino-acid core peptide for α-amanitin or phallacidin, respectively; and a C-terminal 17-amino-acid follower peptide (recognition sequence). The enzymes involved in post-modifications such as multiple hydroxylation, epimerization (for the D-amino acid in phallotoxins), and the formation of tryptathionine, are currently still unknown.

GmPOPB, a prolyl oligopeptidase B from *Galerina marginata* that belongs to classic serine proteases, showed dual function for the proteolytic processing and macrocyclization of the α-amanitin precursor peptide.[40] GmPOPB hydrolyzed at the Pro^{10} residue to cleave the leader peptide and cyclized the C-terminal 25-mer with transpeptidation at the second Pro to produce the cyclic octapeptide (Fig. 5). In the proposed mechanism based on the kinetic studies and the substrate-bound structures of GmPOPB,[40–42] GmPOPB releases the 25-mer peptide after the removal of the leader peptide, and the peptide then rebinds to GmPOPB and undergoes a macrocyclization reaction. A range of peptide substrates containing modified amino acids or naturally occurring cycloamanide sequences have been synthesized and tested. GmPOPB shows a broad substrate profile in macrocyclization.[43]

7. Biosynthesis of omphalotin A

Omphalotin A is a cyclic dodecapeptide isolated from the jack-o'-lantern mushroom *Omphalotus olearius* by Sterner *et al.* in 1997 (Fig. 6A).[44] It shows potent and selective activity against the plant-pathogenic nematode *Meloidogyne incognita*.[44] Omphalotin A and its derivatives contain nine *N*-methylations of the amide backbone and, until recently, were assumed to be products of an NRPS. Velden *et al.* and Ramm *et al.* identified the biosynthetic genes for omphalotin and provided the first example of the incorporation of α-*N*-methyllation in a peptide backbone *via* the RiPP pathway.[45,46] A basic local alignment search tool (BLAST) search of the linearized omphalotin peptide sequence (WVIVVGVIGVIG) against the *O. olearius* genome revealed a gene-encoding protein, OphA[45] (or OphMA[46]), containing a *S*-adenosylmethionine (SAM)-dependent methyltransferase and the omphalotin core sequence containing the C-terminus (Fig. 6B). Four adjacent genes encoding a prolyl oligopeptidase (OphP), an acetyltransferase (OphC), and two P450 monooxygenases (OphM1 and OphM2) were assigned to the omphalotin biosynthesis (*oph*) gene cluster together with OphA.

OphA catalyzes the auto-α-*N*-methylation of the peptide substrate fused to its C-terminus. The full-length OphA recombinant protein was expressed and purified from *E. coli* and analyzed by electrospray ionization time-of-flight MS.[45] After one-day expression, zero to ten methylations were observed on the omphalotin precursor core peptide, with ten methylations being the most common. Up to eleven methylations can be observed after five days of expression. The pattern of methylation was the same as that in the omphalotins, with two additional methylations in the C-terminal recognition sequence. The methylation sites were localized unambiguously with electron-transfer dissociation fragmentation.

The residues Y98 and S129 located within the potential SAM-binding pocket of OphA were substituted by alanine, resulting in the abolishment of methylation on the C-terminal omphalotin core sequence. This suggests the methylation property of the methyltransferase domain of OphA. On the

 This journal is © The Royal Society of Chemistry 2019

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

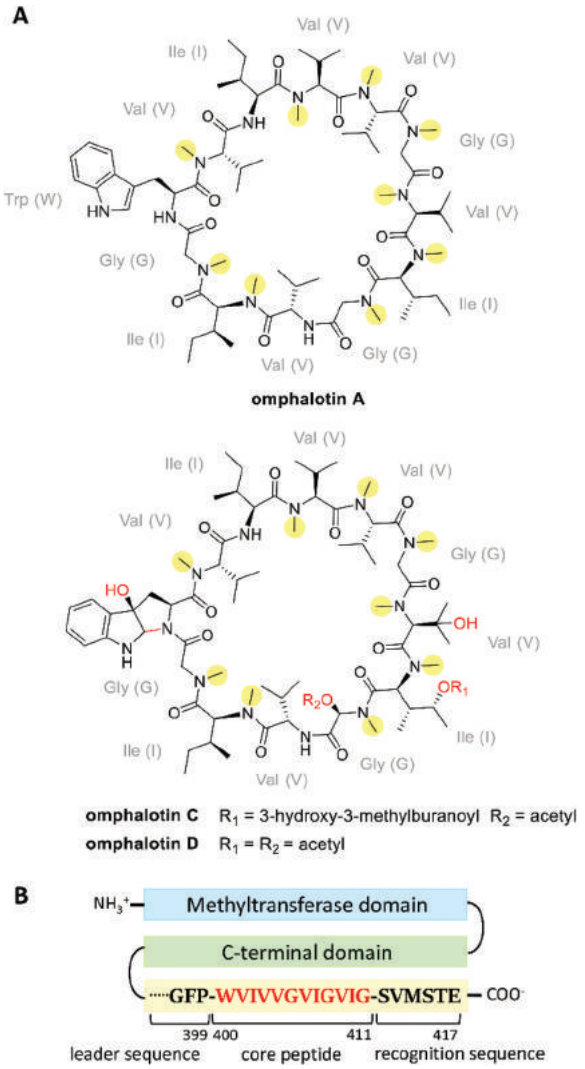

Fig. 6 (A) Structure of omphalotins A, C and D, with α-nitro methyl groups labelled in yellow. (B) Primary structure of OphA protein, which contains a methyltransferase domain (blue), clasp domain (green), and C-terminal substrate peptide (red) flanked by a leader and recognition sequence.

other hand, the automethylation property was retained after the incorporation of a TEV cleavage site (OphA-TEV) *via* the mutation of residues 392 through 396, suggesting the potential to obtain cleavable α-*N*-methylated linear peptides with engineered OphA.[45]

The OphA core peptide was replaced with sequences containing two naturally occurring backbone *N*-methylated therapeutic peptides, cyclosporin A and dictyonamide A. HPLC-MS/MS HCD analysis revealed multiple methylations on the core peptides. Although the methylation patterns were not identical to those of the natural products, the results revealed the ability of OphA to produce alternative core sequences with different extents of methylation.[45]

OphP, a prolyl oligopeptidase, was shown to catalyze the release of *N*-methylated omphalotin core peptide from OphA, as confirmed by the co-expression of OphMA and OphP in the *Pichia pastoris* strain GS115.[45] Other enzymes encoded on the *oph* gene cluster, including an acetyltransferase (OphC) and two P450 monooxygenases (OphM1 and OphM2), were proposed to be involved in the generation of omphalotin derivatives such as omphalotins C and D (Fig. 6A)[47] but have not been characterized. The results of omphalotin biosynthetic studies led to the discovery of a new class of RiPP pathway and an *N*-methytransferase that performs autocatalytic α-*N*-methylation on its C-terminal core sequence.

More recently, the structure of OphA was reported including complexes with cofactors [SAM and *S*-adenosyl homocysteine (SAH)], cofactor analog, substrate, and product.[48] OphA forms a dimer with a complex catenane-like arrangement. The C-terminal substrate peptide is clamped with its amide nitrogen aligned for nucleophilic attack on the methyl group of SAM. The enzyme stabilizes the reaction intermediates with a base-catalyzed mechanism for further α-*N*-methylation.

8. Biosynthesis of strobilurin A

Strobilurin A and oudemansin A (also called mucidin) are antifungal metabolites from various Basidiomycota fungi. They have been important leads for developing β-methoxyacrylate agricultural fungicides such as azoxystrobin and Kresoxim methyl, the most widely used fungicides in the world (Fig. 7). Their structures contain β-methoxyacrylate moieties that block ATP synthesis by targeting the Qo site of complex III of the mitochondrial electron transport chain.[49]

Following the isotope feeding experiments showing that strobilurin A is derived from an unusual benzoate starter unit formed by the degradation of phenylalanine *via* cinnamate,[50] the strobilurin A biosynthetic gene clusters were identified from two Basidiomycota fungi, *Strobilurus tenacellus* and *Strobilurus lutea*.[51] StPKS1, a highly reducing iterative fungal polyketide synthase (PKS) from *S. tenacellus*, contains an unusual partial αβ-hydrolase domain and a SAM-dependent *C*-methyltransferase domain downstream of the usually terminal ACP domain. Benzoyl CoA is used as a rare start unit for StPKS1 and converted into a highly unusual *E*,*Z*,*E*-triene, prestrobilurin A (Fig. 8). Based on heterologous expression experiments in *Aspergillus oryzae* NSAR1, the genes *str11*, *str8* and *str10*, which respectively encode a phenylalanine ammonia lyase, a non-haem iron oxygenase and a CoA ligase, were shown to be effective sources of benzoyl CoA *in vivo*. Str9, a FAD-dependent oxygenase, is responsible for the oxidative

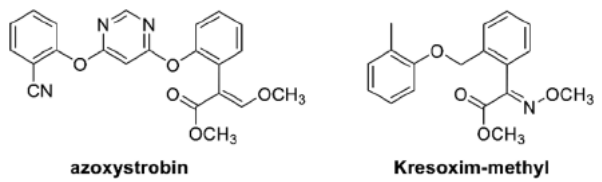

Fig. 7 Structure of fungicides, azoxystrobin and Kresoxim-methyl.

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

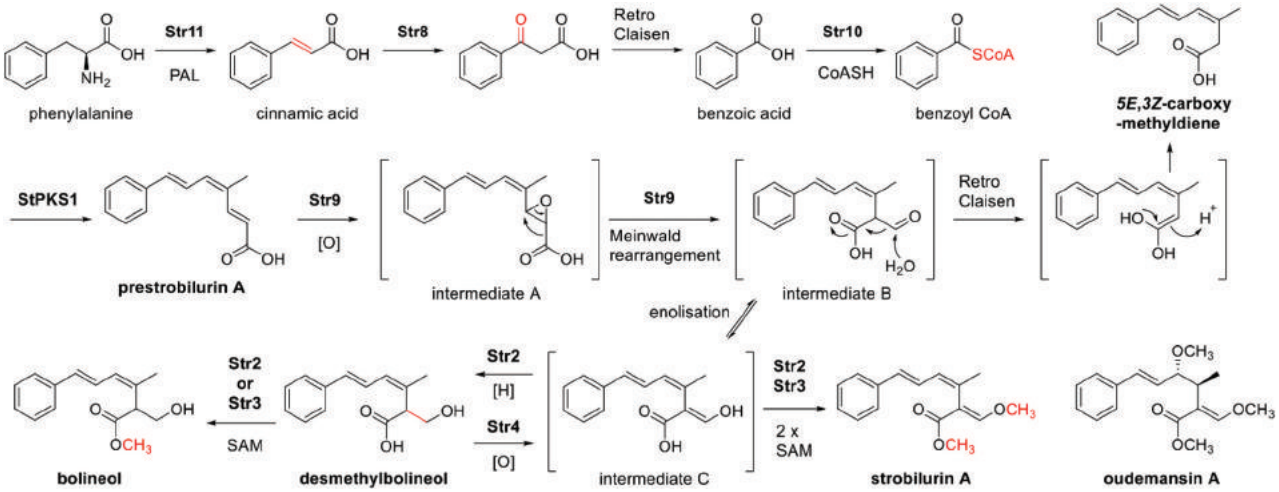

Fig. 8 Biosynthesis of strobilurin A.

rearrangement of prestrobilurin to form the key β-methoxyacrylate toxophore of the strobilurin fungicides. This is supported by both *in vivo* and *in vitro* experiments, in which the rearranged product intermediate B was observed by LCMS after the incubation of a mixture of prestrobilurin, Str9, and NADH/NADPH. The proposed mechanism involves the epoxidation of the 2,3-olefin of prestrobilurin A followed by the Meinwald rearrangement of intermediate A to give the aldehyde intermediate B, which appears to easily undergo rapid retro-Claisen reaction to give 5*E*,3*Z*-carboxy-methyldiene.

Str2 encoding short-chain dehydrogenase and Str4 encoding glucose-methane-choline oxidase are suggested to function in opposite ways to reduce aldehydes and oxidize primary alcohols into aldehydes, respectively. The presence of Str2 or Str4 in the heterologous system resulted in the production of bolineol or strobilurin A, respectively. The rapid enolization of intermediate B gives the proposed β-methoxyacrylate intermediate C, and methylation catalyzed by Str2 or Str3 methyltransferase gives strobilurin A.

9. Conclusion and perspective

In addition to playing important roles in carbon recycling and ecosystem functioning, Basidiomycota fungi are known to exhibit biological and pharmacological properties.[52] As described above, cumulative numbers of biosynthetic pathways for the construction of pharmacologically and agriculturally important natural products from Basidiomycota have been reported in recent years. Characterized enzymes with novel and unique functions, such as the UbiA homolog EriG involved in constructing the cyathane diterpenoid skeleton and the methyltransferase OphA involved in omphalotin biosynthesis, will be added to the database of enzymes, thus facilitating the accurate prediction and annotation of biosynthetic gene clusters.

Transcriptomics and alternative heterologous expression hosts can be useful tools in the efficient characterization of biosynthetic genes in Basidiomycota. The genes in these fungi usually contain many intron-rich sequences, making it difficult for accurate gene annotation and prediction. The availability of transcriptomic data can correct coding sequences and refine the procedure for gene annotation. On the other hand, the phenomena of non-cluster genes involved in the biosynthesis of Basidiomycota natural products such as triterpenoids have been investigated.[28] The combination of transcriptomic analysis and metabolic analysis of fungal tissues collected at different stages of the life cycle under laboratory growth conditions or outdoor sampling will allow better prediction of promising biosynthetic gene candidates. Since most Basidiomycota fungi are genetically intractable, the development of alternative heterologous expression hosts is required for enzyme expression and biosynthetic pathway reconstitution. In addition to the more well-established systems for the functional characterization of enzymes, including *E. coli*, *Saccharomyces cerevisiae*, and *Pichia pastoris*, some fungal heterologous hosts such as *Aspergillus oryzae* have been shown to be alternative systems for pathway assembly and reconstitution, as exemplified in the aforementioned studies. The development of a Basidiomycota heterologous host could allow protein refolding and post-modification of Basiomycota biosynthetic genes. The model basidiomycete *Coprinopsis cinerea* was shown to produce aryl-alcohol oxidases that are involved in the degradation of lignin from white-rot fungus *Pleurotus sapidus*, and their enzymatic activities reached up to 125 U L^{-1}.[53] In contrast, when expressed in *Aspergillus nidulans* and upon post-modification in *E. coli* and *S. cerevisiae*, the aryl-alcohol oxidases resulted in low yields. Furthermore, the application of modern genetic tools such as CRISPR-Cas9 enables site-specific and efficient gene integration in both host producing strain manipulation and heterologous expression systems.[54,55]

In conclusion, the mechanisms of the biosynthesis of many chemical constituents in Basidiomycota remain elusive. The above examples show that they encode a wealth of novel bio-

 This journal is © The Royal Society of Chemistry 2019

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

synthetic enzymes and are currently underexplored. Improving synthetic biology and chemical methods will contribute to accessing the structural diversity of Basidiomycota natural products. With the notable biological activities and medicinal properties of natural products from Basidiomycota fungi, the exploration of the chemoenzymatic tools utilized in biosynthesis could accelerate the development of these important molecules for pharmacological and agricultural applications.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work was supported by the Ministry of Science and Technology (MOST), Taiwan (106-2113-M-001-008-MY2) and the Institute of Biological Chemistry, Academia Sinica. We thank Li-Chen Lin for helpful discussion.

References

1 J.-J. Zhong and J.-H. Xiao, *Biotechnology in China I: From Bioreaction to Bioseparation and Bioremediation*, Springer, 2009, pp. 79–150.
2 H.-P. Chen and J.-K. Liu, *Progress in the Chemistry of Organic Natural Products*, Springer, 2017, pp.1–201.
3 I. V. Grigoriev, R. Nikitin, S. Haridas, A. Kuo, R. Ohm, R. Otillar, R. Riley, A. Salamov, X. L. Zhao, F. Korzeniewski, T. Smirnova, H. Nordberg, I. Dubchak and I. Shabalov, *Nucleic Acids Res.*, 2014, **42**, D699–D704.
4 M. B. Quin, C. M. Flynn and C. Schmidt-Dannert, *Nat. Prod. Rep.*, 2014, **31**, 1449–1473.
5 C. Schmidt-Dannert, *Adv. Biochem. Eng./Biotechnol.*, 2015, **148**, 19–61.
6 H. Xiao and J. J. Zhong, *Trends Biotechnol.*, 2016, **34**, 242–255.
7 C. M. Flynn and C. Schmidt-Dannert, *Appl. Environ. Microbiol.*, 2018, **84**, e00036-18.
8 F. Kavanagh, A. Hervey and W. J. Robbins, *Proc. Natl. Acad. Sci. U. S. A.*, 1951, **37**, 570–574.
9 C. Davidovich, A. Bashan, T. Auerbach-Nevo, R. D. Yaggie, R. R. Gontarek and A. Yonath, *Proc. Natl. Acad. Sci. U. S. A.*, 2007, **104**, 4291–4296.
10 S. M. Poulsen, M. Karlsson, L. B. Johansson and B. Vester, *Mol. Microbiol.*, 2001, **41**, 1091–1099.
11 R. S. Daum, S. Kar and P. Kirkpatrick, *Nat. Rev. Drug Discovery*, 2007, **6**, 865–866.
12 H. Hasler, *Neue aspekte der biosynthese von pleuromutilin*, PhD thesis, ETH Zurich, 1979.
13 T. Tsukagoshi, T. Tokiwano and H. Oikawa, *Biosci., Biotechnol., Biochem.*, 2007, **71**, 3116–3121.
14 A. M. Bailey, F. Alberti, S. Kilaru, C. M. Collins, K. de Mattos-Shipley, A. J. Hartley, P. Hayes, A. Griffin, C. M. Lazarus, R. J. Cox, C. L. Willis, K. O'Dwyer, D. W. Spence and G. D. Foster, *Sci. Rep.*, 2016, **6**, 25202.
15 F. Alberti, K. Khairudin, E. R. Venegas, J. A. Davies, P. M. Hayes, C. L. Willis, A. M. Bailey and G. D. Foster, *Nat. Commun.*, 2017, **8**, 1831.
16 M. Yamane, A. Minami, C. Liu, T. Ozaki, I. Takeuchi, T. Tsukagoshi, T. Tokiwano, K. Gomi and H. Oikawa, *ChemBioChem*, 2017, **18**, 2317–2322.
17 M. M. Xu, M. R. Jia, Y. J. Hong, X. H. Yin, D. J. Tantillo, P. J. Proteau and R. J. Peters, *Org. Lett.*, 2018, **20**, 1200–1202.
18 J. A. Enquist and B. M. Stoltz, *Nat. Prod. Rep.*, 2009, **26**, 661–680.
19 W. A. Ayer and H. Taube, *Tetrahedron Lett.*, 1972, **19**, 1917–1920.
20 T. Tsai-Teng, C. Chin-Chu, L. Li-Ya, C. Wan-Ping, L. Chung-Kuang, S. Chien-Chang, H. F. Chi-Ying, C. Chien-Chih and Y. J. Shiao, *J. Biomed. Sci.*, 2016, **23**.
21 Y. l. Yang, S. Zhang, K. Ma, Y. Xu, Q. Tao, Y. Chen, J. Chen, S. Guo, J. Ren and W. Wang, *Angew. Chem., Int. Ed.*, 2017, **129**, 4827–4830.
22 H. C. Lin, Y. H. Chooi, S. Dhingra, W. Xu, A. M. Calvo and Y. Tang, *J. Am. Chem. Soc.*, 2013, **135**, 4616–4619.
23 M. J. Smanski, Z. Yu, J. Casper, S. Lin, R. M. Peterson, Y. Chen, E. Wendt-Pienkowski, S. R. Rajski and B. Shen, *Proc. Natl. Acad. Sci. U. S. A.*, 2011, **108**, 13498–13503.
24 S. Baby, A. J. Johnson and B. Govindan, *Phytochemistry*, 2015, **114**, 66–101.
25 M. C. Lu, M. El-Shazly, T. Y. Wu, Y. C. Du, T. T. Chang, C. F. Chen, Y. M. Hsu, K. H. Lai, C. P. Chiu, F. R. Chang and Y. C. Wu, *Pharmacol. Ther.*, 2013, **139**, 124–156.
26 J. L. Rios, I. Andujar, M. C. Recio and R. M. Giner, *J. Nat. Prod.*, 2012, **75**, 2016–2044.
27 I. Abe, M. Rohmer and G. D. Prestwich, *Chem. Rev.*, 1993, **93**, 2189–2206.
28 S. L. Chen, J. Xu, C. Liu, Y. J. Zhu, D. R. Nelson, S. G. Zhou, C. F. Li, L. Z. Wang, X. Guo, Y. Z. Sun, H. M. Luo, Y. Li, J. Y. Song, B. Henrissat, A. Levasseur, J. Qian, J. Q. Li, X. Luo, L. C. Shi, L. He, L. Xiang, X. L. Xu, Y. Y. Niu, Q. S. Li, M. V. Han, H. X. Yan, J. Zhang, H. M. Chen, A. P. Lv, Z. Wang, M. Z. Liu, D. C. Schwartz and C. Sun, *Nat. Commun.*, 2012, **3**, 913.
29 W. F. Wang, H. Xiao and J. J. Zhong, *Biotechnol. Bioeng.*, 2018, **115**, 1842–1854.
30 C. Andersson, J. Kristinsson and J. Gry, *Occurrence and use of hallucinogenic mushrooms containing psilocybin alkaloids*, Nordic Council of Ministers, 2008.
31 A. Hofmann, R. Heim, A. Brack, H. Kobel, A. Frey, H. Ott, T. Petrzilka and F. Troxler, *Helv. Chim. Acta*, 1959, **42**, 1557–1572.
32 A. Hofmann, R. Heim, A. Brack and H. Kobel, *Experientia*, 1958, **14**, 107–109.
33 F. Hasler, U. Grimberg, M. A. Benz, T. Huber and F. X. Vollenweider, *Psychopharmacology*, 2004, **172**, 145–156.

Published on 14 December 2018. Downloaded by Icahn School of Medicine at Mount Sinai on 2/3/2019 10:44:37 PM.

34 J. Fricke, F. Blei and D. Hoffmeister, *Angew. Chem., Int. Ed.*, 2017, **56**, 12352–12355.
35 F. Blei, F. Baldeweg, J. Fricke and D. Hoffmeister, *Chem. – Eur. J.*, 2018, **24**, 10028–10031.
36 J. Walton, *The Cyclic Peptide Toxins of Amanita and Other Poisonous Mushrooms*, Springer, 2018.
37 T. Wieland, *Int. J. Pept. Protein Res.*, 1983, **22**, 257–276.
38 H. E. Hallen, H. Luo, J. S. Scott-Craig and J. D. Walton, *Proc. Natl. Acad. Sci. U. S. A.*, 2007, **104**, 19097–19101.
39 P. Zhou, K. A. Silverstein, L. Gao, J. D. Walton, S. Nallu, J. Guhlin and N. D. Young, *BMC Bioinf.*, 2013, **14**, 335.
40 H. Luo, S.-Y. Hong, R. M. Sgambelluri, E. Angelos, X. Li and J. D. Walton, *Chem. Biol.*, 2014, **21**, 1610–1617.
41 C. M. Czekster, H. Ludewig, S. A. McMahon and J. H. Naismith, *Nat. Commun.*, 2017, **8**, 1045.
42 C. M. Czekster and J. H. Naismith, *Biochemistry*, 2017, **56**, 2086–2095.
43 R. M. Sgambelluri, M. O. Smith and J. D. Walton, *ACS Synth. Biol.*, 2018, **7**, 145–152.
44 A. Mayer, M. Kilian, B. Hoster, O. Sterner and H. Anke, *Pestic. Sci.*, 1999, **55**, 27–30.
45 N. S. van der Velden, N. Kalin, M. J. Helf, J. Piel, M. F. Freeman and M. Kunzler, *Nat. Chem. Biol.*, 2017, **13**, 833–835.
46 S. Ramm, B. Krawczyk, A. Muhlenweg, A. Poch, E. Mosker and R. D. Sussmuth, *Angew. Chem., Int. Ed.*, 2017, **56**, 9994–9997.
47 E. Buchel, U. Martini, A. Mayer, H. Anke and O. Sterner, *Tetrahedron*, 1998, **54**, 5345–5352.
48 H. Song, N. S. van der Velden, S. L. Shiran, P. Bleiziffer, C. Zach, R. Sieber, A. S. Imani, F. Krausbeck, M. Aebi, M. F. Freeman, S. Riniker, M. Kunzler and J. H. Naismith, *Sci. Adv.*, 2018, **4**, eaat2720.
49 W. F. Becker, G. Vonjagow, T. Anke and W. Steglich, *FEBS Lett.*, 1981, **132**, 329–333.
50 Z. Iqbal, L. C. Han, A. M. Soares-Sello, R. Nofiani, G. Thormann, A. Zeeck, R. J. Cox, C. L. Willis and T. J. Simpson, *Org. Biomol. Chem.*, 2018, **16**, 5524–5532.
51 R. Nofiani, K. de Mattos-Shipley, K. E. Lebe, L. C. Han, Z. Iqbal, A. M. Bailey, C. L. Willis, T. J. Simpson and R. J. Cox, *Nat. Commun.*, 2018, **9**, 3949.
52 C. Schmidt-Dannert, *Curr. Opin. Chem. Biol.*, 2016, **31**, 40–49.
53 I. Galperin, A. Javeed, H. Luig, G. Lochnit and M. Ruhl, *Appl. Microbiol. Biotechnol.*, 2016, **100**, 8021–8030.
54 S. S. Sugano, H. Suzuki, E. Shimokita, H. Chiba, S. Noji, Y. Osakabe and K. Osakabe, *Sci. Rep.*, 2017, **7**, 1260.
55 H. Qin, H. Xiao, G. Zou, Z. H. Zhou and J. J. Zhong, *Process Biochem.*, 2017, **56**, 57–61.

 This journal is © The Royal Society of Chemistry 2019